Total Points: [14] [U1HW7 - W10 - Robot Hand]

1. What key skill are students practicing when they assign roles, share responsibilities, and resolve disagreements?

- Competitive arguing
- Advanced calculus
- Memorization of facts
- Collaboration and teamwork
- Silent reading
- Handwriting speed

Answer:

2. What component of the human hand acts like a string to pull the fingers toward the palm?

- The fulcrum
- The lever
- The wrist
- The skin
- The tendons
- The joints

Answer:

3. Which anatomical component connects to the tendons and powers them?

- The joints
- The skin
- The wrist
- The fulcrum
- The bones
- The muscle

Answer:

4. How long must the designed hand hold an item to meet the stated testable goals?

- 20 seconds
- 1 minute
- 10 seconds
- 30 seconds
- 5 seconds
- 2 seconds

Answer:

5. Why is the "Iteration & Redesign" step of the engineering process important for students?

• To make the project take as long as possible

• To copy exactly what the teacher built

• To learn from failures and improve the performance of the design

• To prove that the first attempt is always perfect

• To make the robotic hand look more colorful

• To use up all the remaining glue and tape

Answer:

6. Which part of the human hand provides enough friction so that gripped items do not slip away?

• The knuckles

• The joints

• The bones

• The wrist bone

• The muscles

• The skin

Answer:

7. Why is it valuable for engineers to design low-cost robotic hands instead of only expensive ones?

• To make helpful technology accessible to people who cannot afford thousands of dollars

• To make the hands heavier and harder to use

• To make sure only wealthy people can use them

• To decrease the quality so the hand breaks faster

• To stop people from using robotic hands

• To avoid using modern materials

Answer:

8. Which term refers to how the anatomy of a real human hand influences the design?

• Bio-mimicry Connection

• Accessibility & Cost

• Societal Impact

• Success Criteria Analysis

• Group Dynamics

• Critical Failures

Answer:

9. A lever is used to give the user which specific benefit?

- Thermal insulation
- Electrical conductivity
- Reduced gravity
- Mechanical advantage
- Increased friction
- A lighter load

Answer:

10. Where is the effort located in a 3rd class lever?

- Behind the pivot point
- In the middle
- Mixed with the load
- On the far left end only
- Underneath the fulcrum
- On the far right end only

Answer:

11. How is the "fulcrum" defined in relation to a lever?

- The rigid bar connecting the parts
- The object being lifted or moved
- The place where the lever or levers are free to rotate
- The result of mechanical advantage
- The place where you apply force
- The weight of the load

Answer:

12. Why are robotic grippers necessary for industries like space exploration or handling hazardous waste?

- Because robots look cooler than humans in photos
- Because robots get lonely in space
- To use up extra electricity
- To prevent humans from having jobs
- To ensure the mission costs more money
- To perform tasks that are too dangerous or deadly for humans to do directly

Answer:

Answer:

13. Which class of lever has the load situated in the middle?

- Compound lever
- 4th class lever
- 2nd class lever
- 3rd class lever
- Reverse lever
- 1st class lever

Answer:

14. What is the primary benefit of studying "bio-mimicry" (how the human hand works) before building the robot?

- To determine if the robot needs fingernails
- To understand how nature creates movement and grip to solve engineering problems
- To see if human skin is waterproof
- To prove that robots are stronger than humans
- To count how many bones are in the wrist
- To learn how to draw a hand for art class